



Chapter 6

Sequential logic

朱晓雷

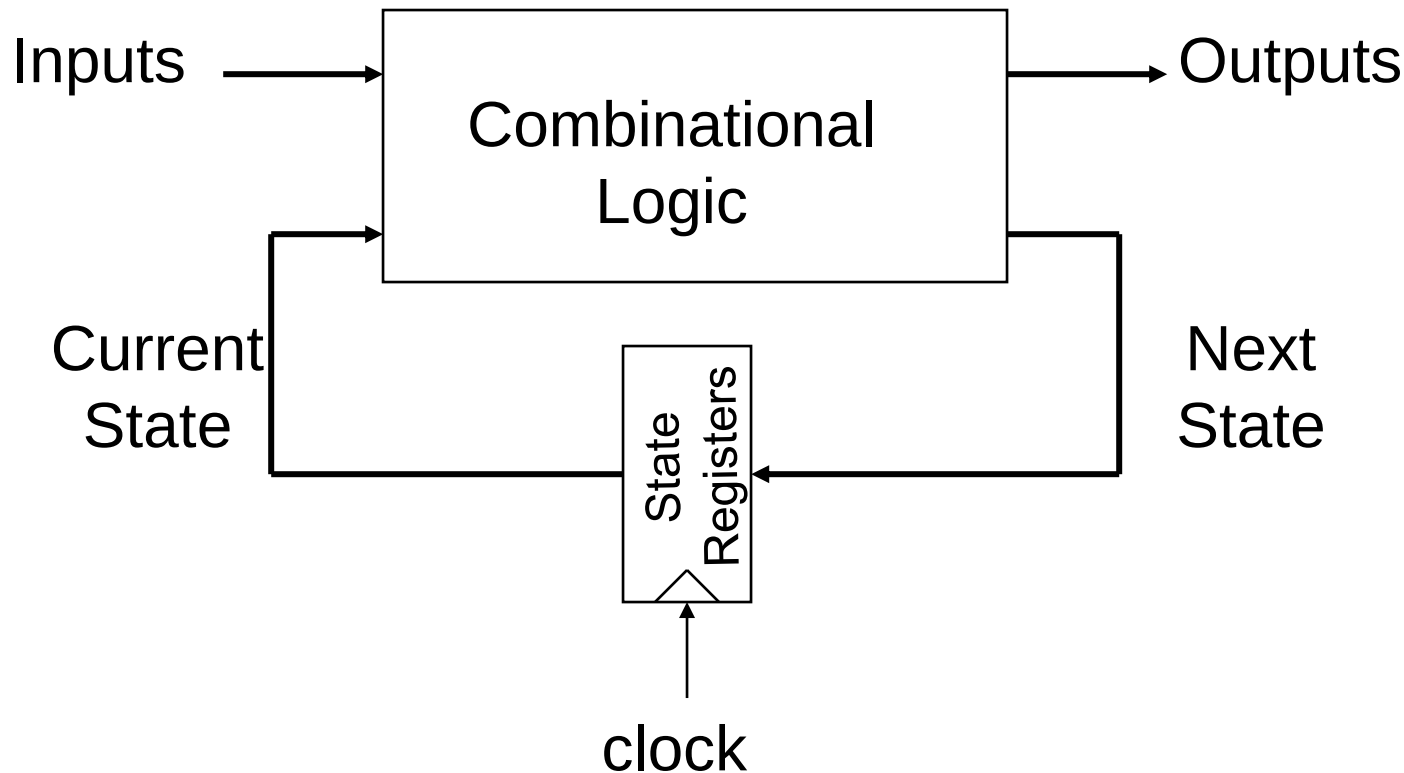
Xl_zhu@zju.edu.cn

2024

*Some Contents Adapted from Copyright Prentice Hall/Pearson , Rabaey's Digital Integrated Circuits, Mary Jane Irwin's VLSI Digital Circuits and University of Bolton, Weste's CMOS VLSI Design



Sequential Logic



- ❑ Sequential machines includes combinational logic and memory elements;
- ❑ Only synchronous timing circuits is discussed in this course.



Memory elements

- The most important components of sequential machines.
- Stores a value controlled by clock.
- In CMOS, memory is created by:
 - capacitance (dynamic);
 - Feedback, refresh (static).



Difference of types of memory elements

- Formats of clock: level or edge
- Maintain the stored value: static or dynamic
- Output related with input: latch or Register (flip-flop)
- Multiple phase: master-slave



Memory element terminology

- Latch: level sensitive; transparent when internal memory is being set from input (Output is changed by input instantly).
- Register (Flip-flop) : edge triggered; not transparent—reading input and changing output are separate events.
- * Flip-flop is defined as a bistable memory element based on cross coupled gates.(in this book)

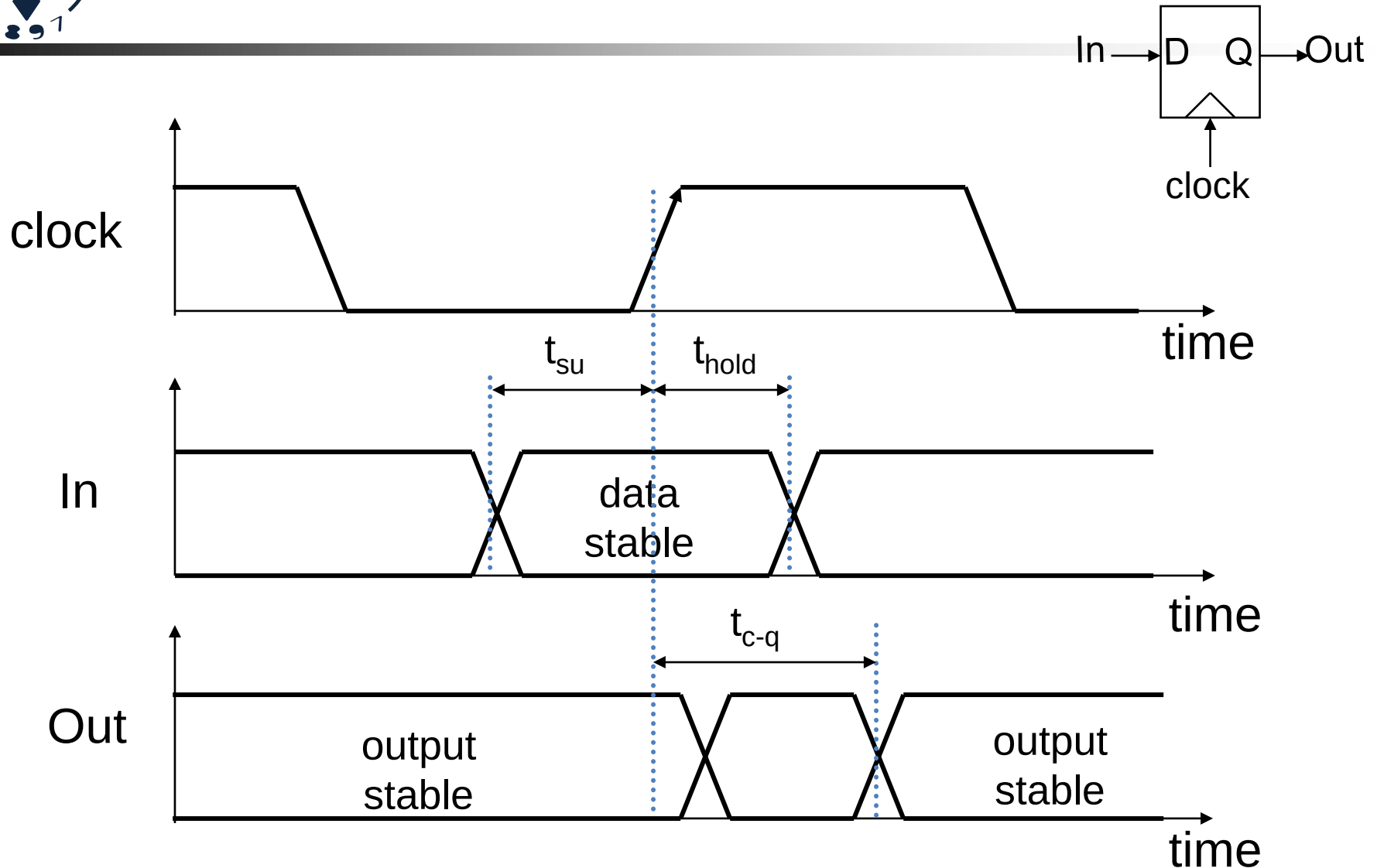


Clock terminology

- Clock edge: rising or falling transition.
- Duty cycle: fraction of clock period for which clock is active (e.g., for active-low clock, fraction of time clock is 0).

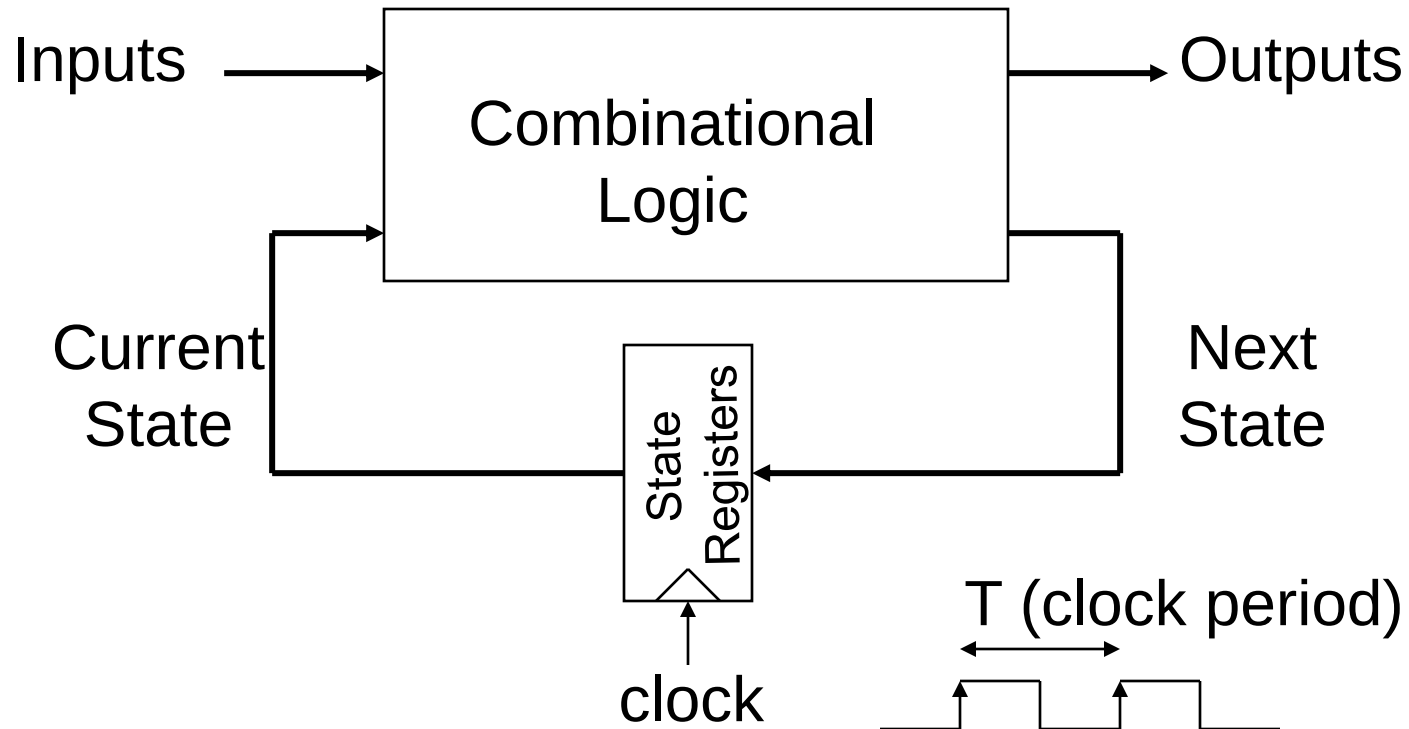


Timing Definitions





System Timing Constraints



$$t_{cdreg} + t_{cdlogic} \geq t_{hold}$$

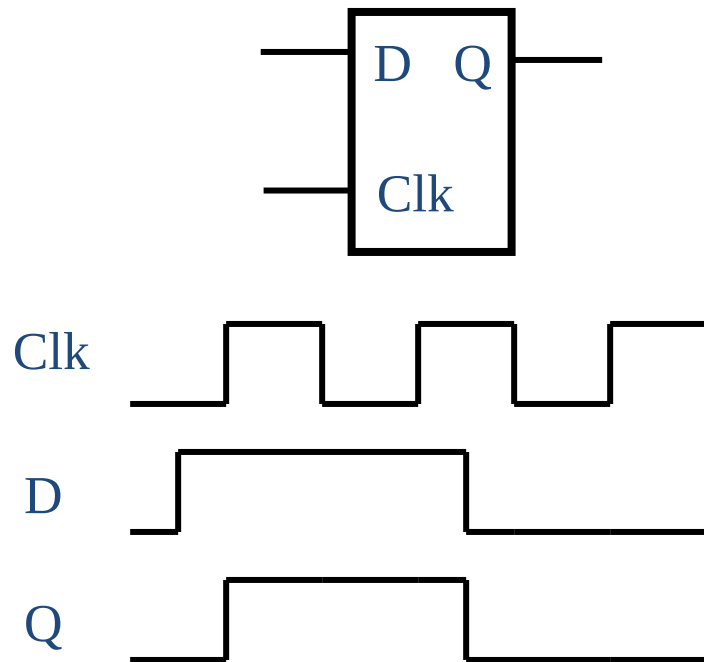
contamination delay - minimum delay of
the combinational logic or register

$$T \geq t_{c-q} + t_{plogic} + t_{su}$$

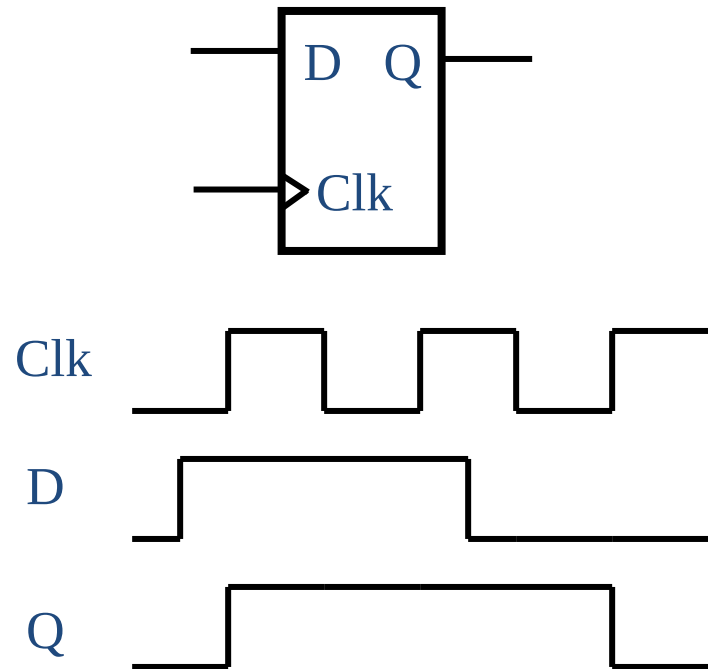


Latch versus Register

- ❑ Latch
stores data when
clock is High



- ❑ Register
stores data when
clock rises

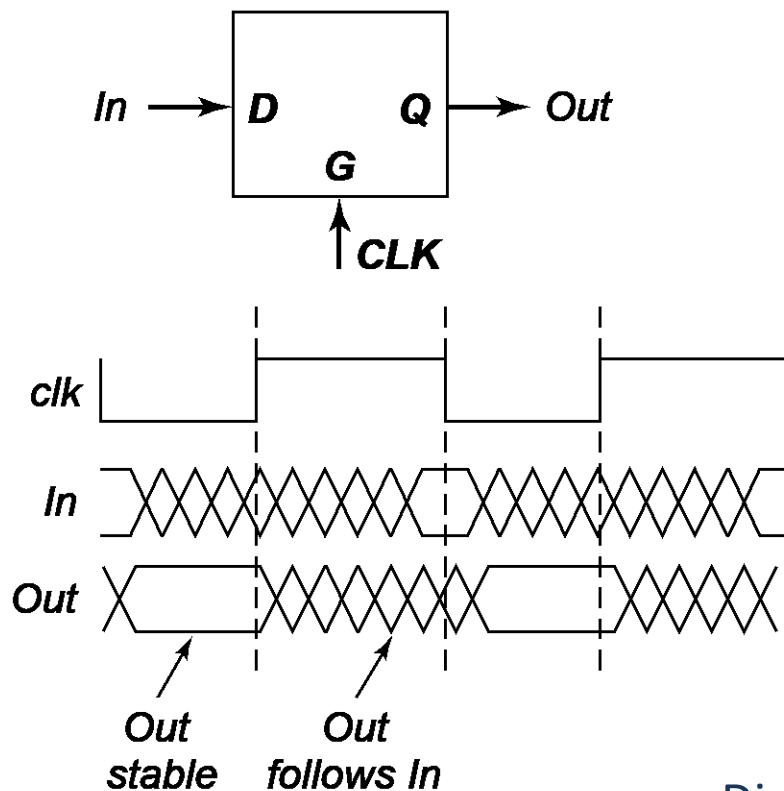




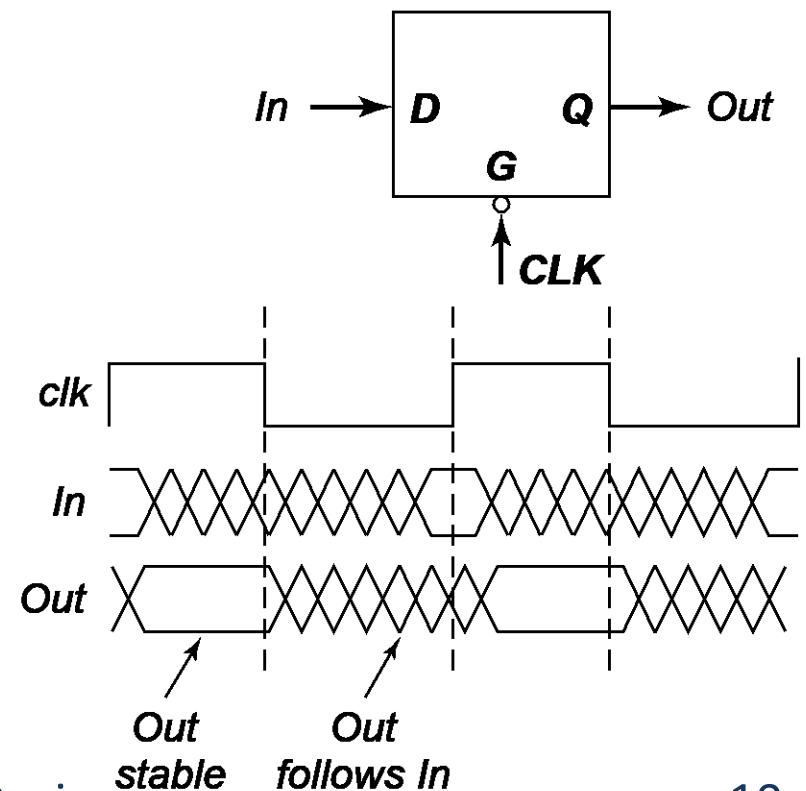
Latches

- **level sensitive** circuit that passes inputs to Q when the clock is high - **transparent** mode
- input sampled on the falling edge of the clock is held stable when clock is low - **hold** mode

Positive Latch

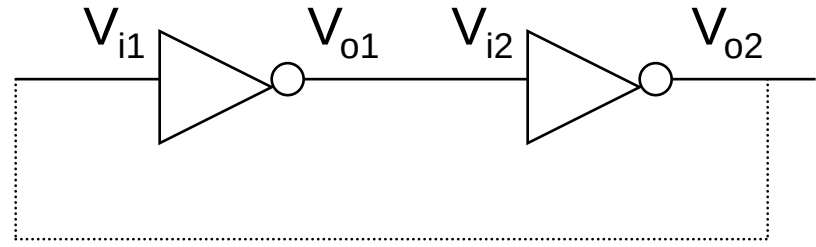
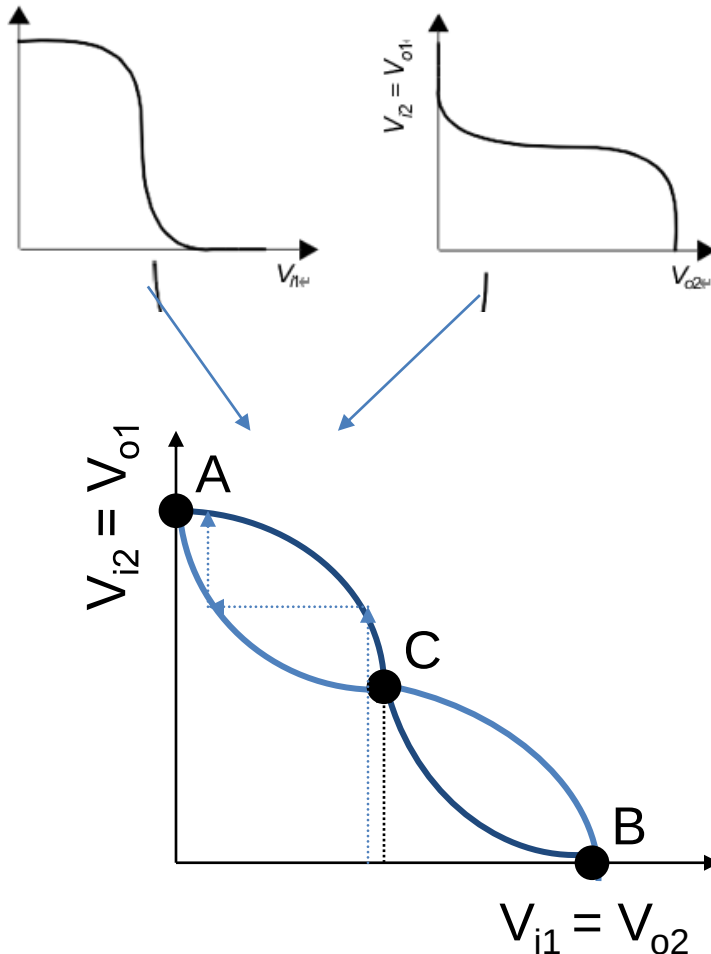


Negative Latch





The Regenerative Property

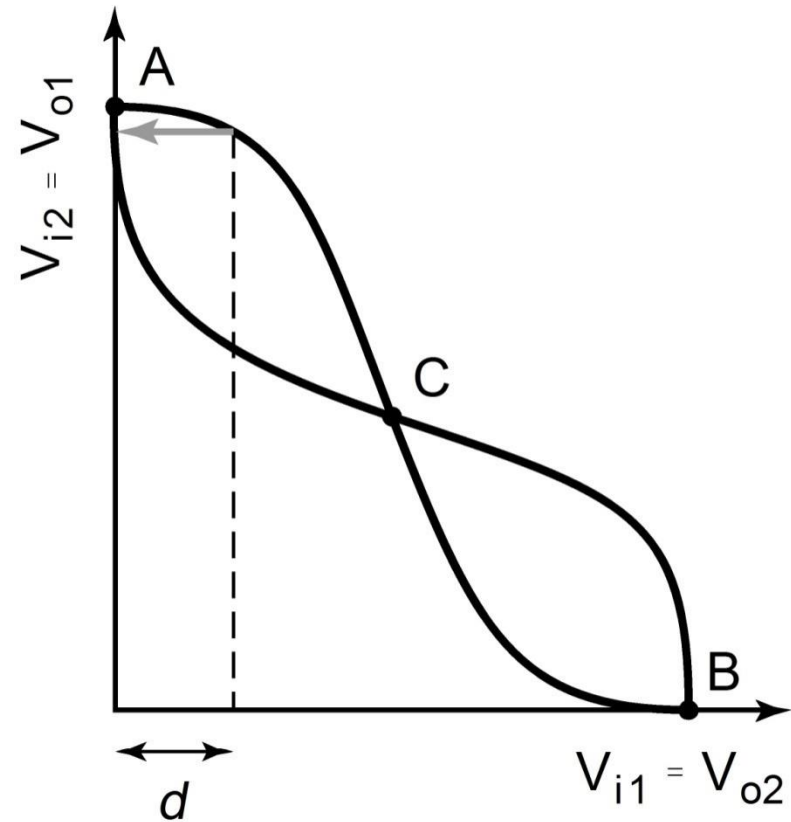
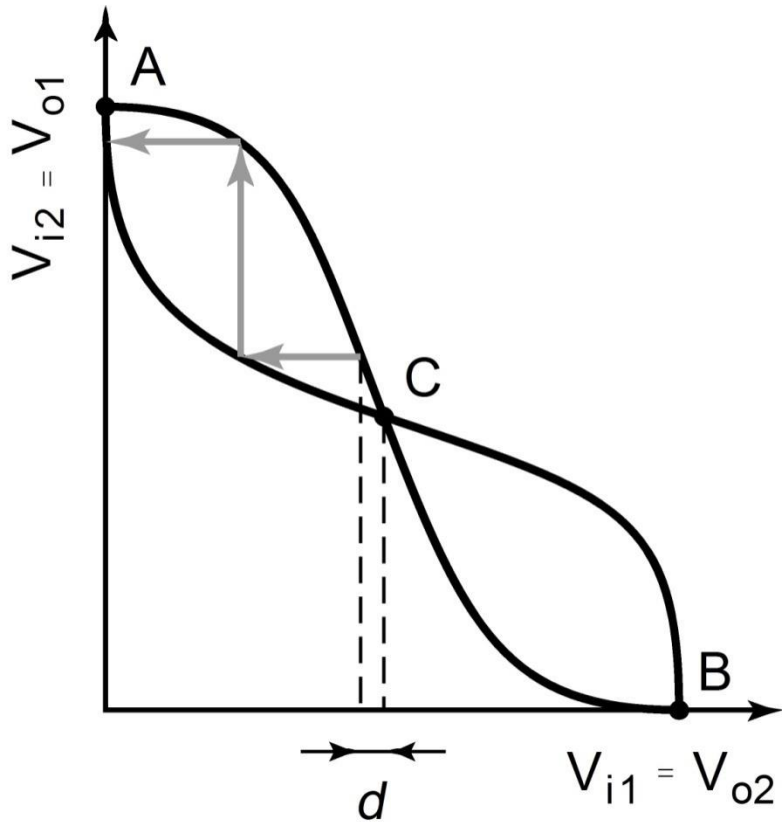


cascaded inverters

If the gain in the transient region is larger than 1, only A and B are stable operation points. C is a metastable operation point.



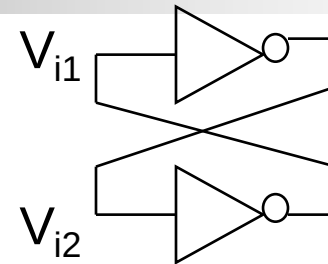
Meta-Stability



Gain should be larger than 1 in the transition region



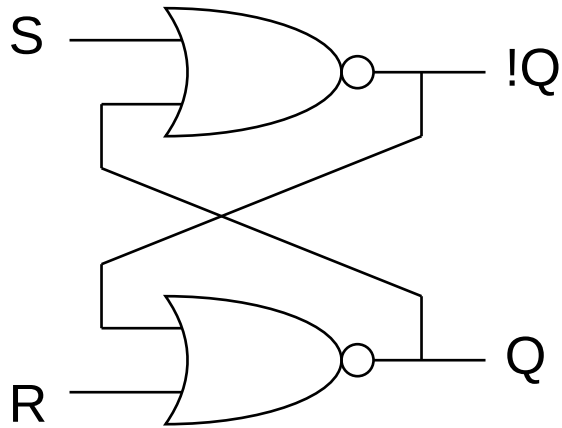
Bistable Circuits



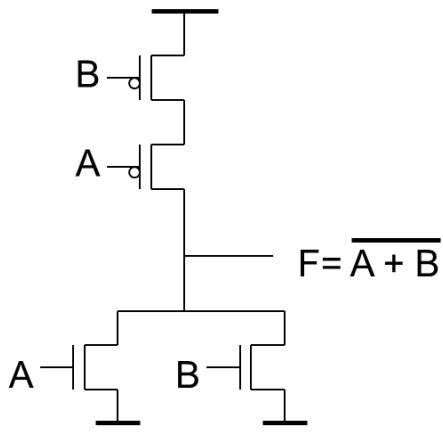
- The cross-coupling of two inverters results in a **bistable circuit** (a circuit with two stable states)
- Have to be able to change the stored value by making A (or B) temporarily unstable by increasing the loop gain to a value larger than 1
 - done by applying a trigger pulse at V_{i1} or V_{i2}
 - the width of the trigger pulse need be only a little larger than the total propagation delay around the loop circuit (twice the delay of an inverter)
- Two approaches used
 - cutting the feedback loop (mux based latch)
 - overpowering the feedback loop (as used in SRAMs)



SR Flip-Flops



S	R	Q	!Q	
0	0	Q	!Q	memory
1	0	1	0	set
0	1	0	1	reset
1	1	0	0	disallowed

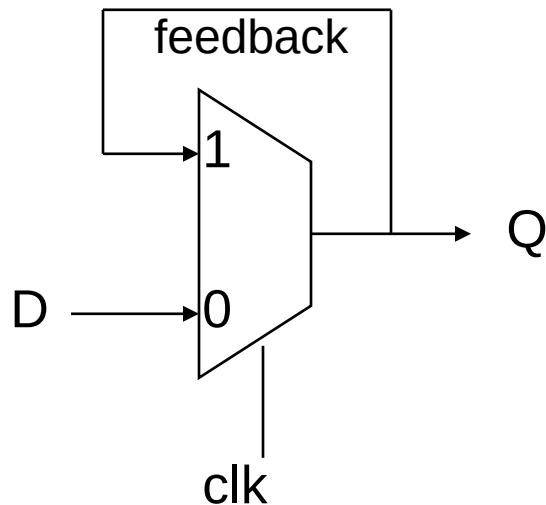


A	B	F
0	0	1
0	1	0
1	0	0
1	1	0



MUX Based Latches

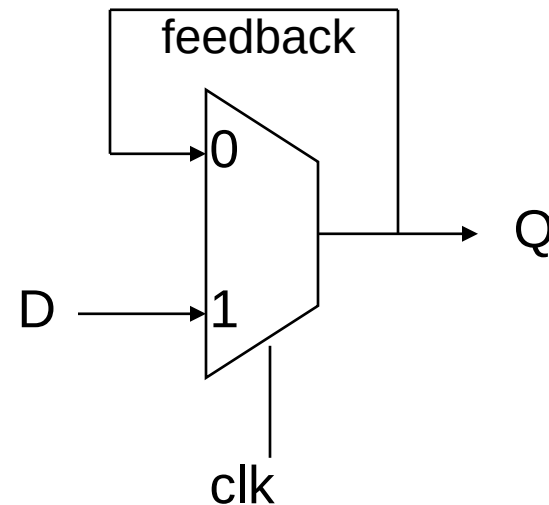
- Change the stored value by cutting the feedback loop



Negative Latch

$$Q = \text{clk} \& Q \mid \neg \text{clk} \& D$$

transparent when the
clock is low



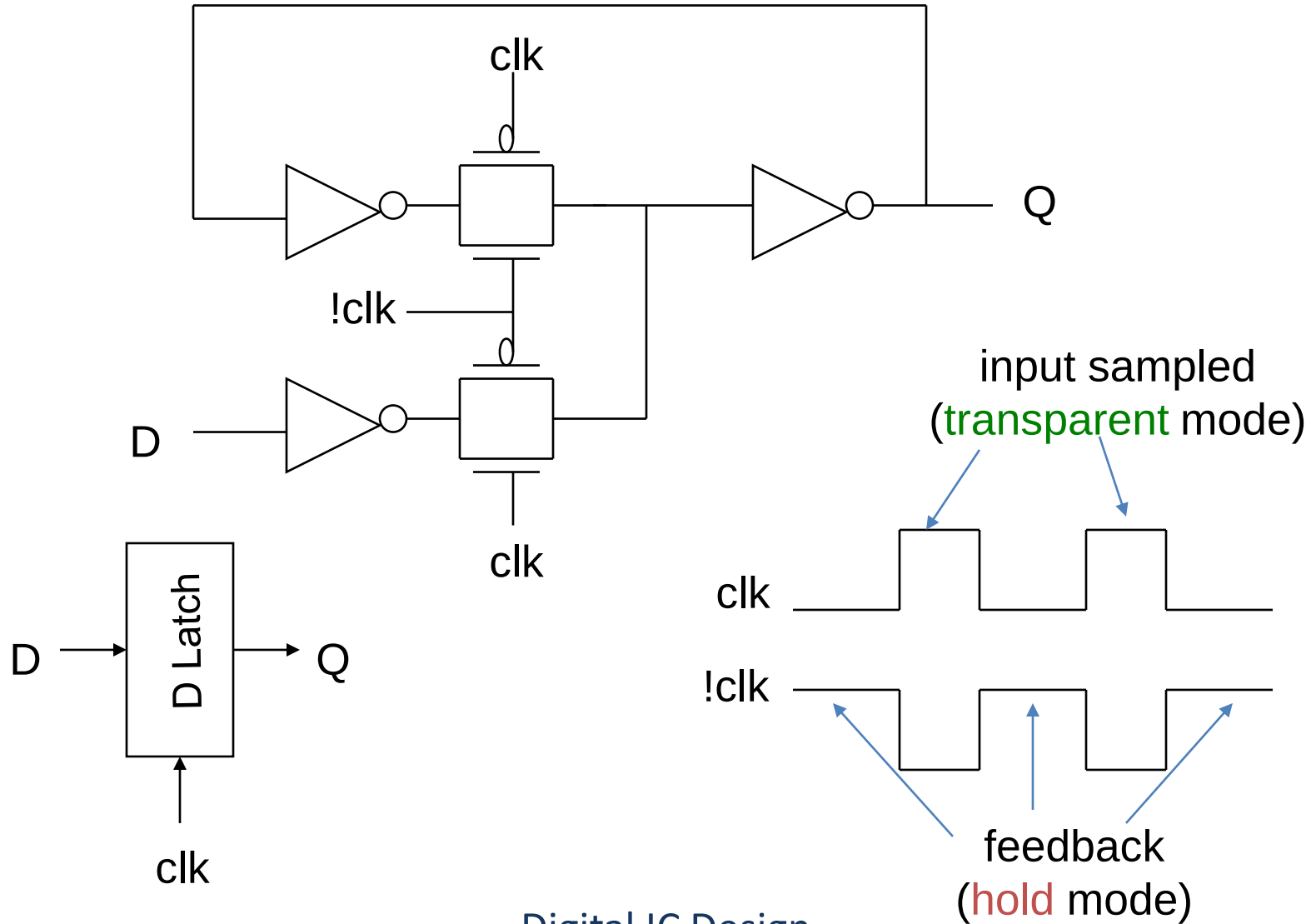
Positive Latch

$$Q = \neg \text{clk} \& Q \mid \text{clk} \& D$$

transparent when the
clock is high

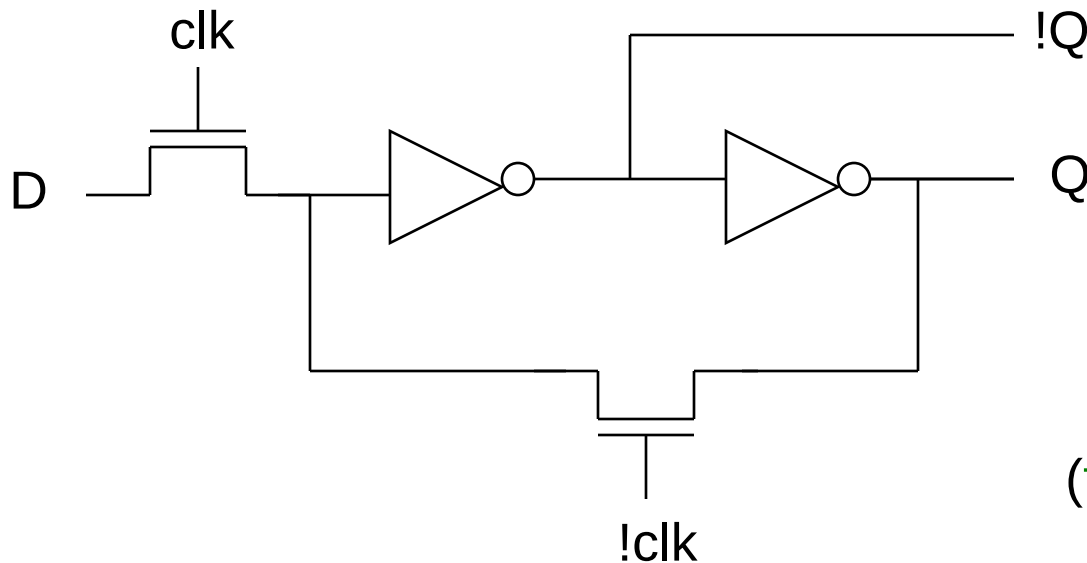


TG MUX Based Latch Implementation

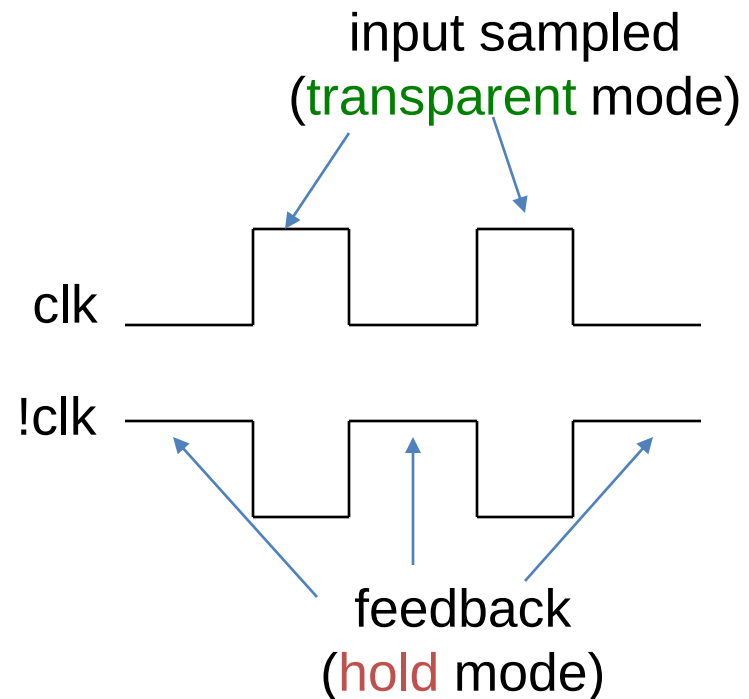




PT (pass transistor) MUX Based Latch Implementation

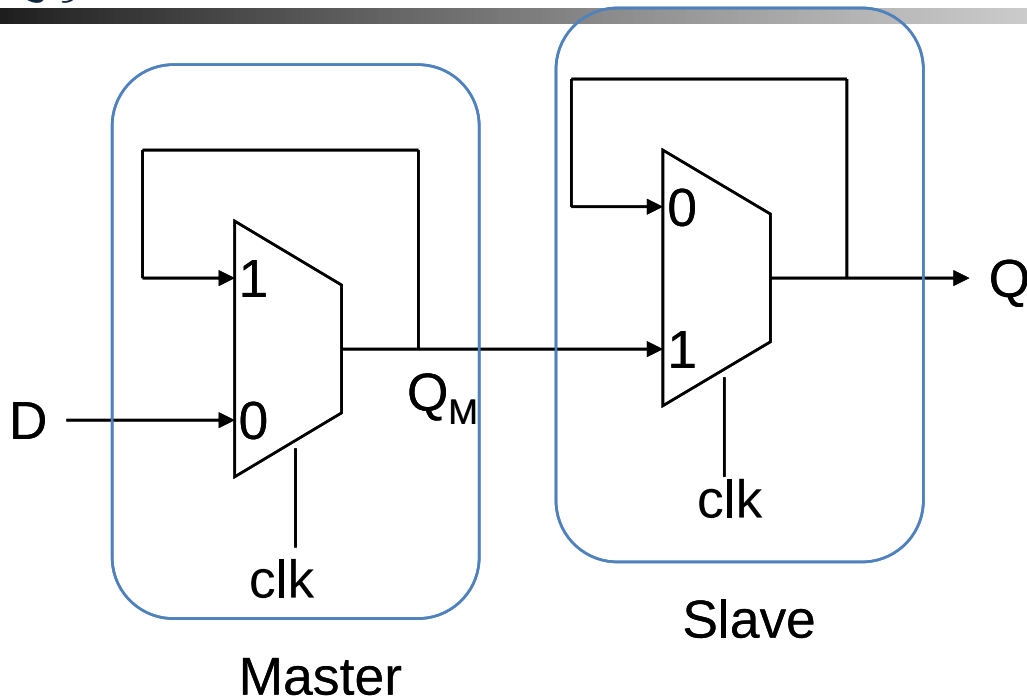


- Reduced clock load, but threshold drop at output of pass transistors so reduced noise margins and performance





Master Slave Based ET Flipflop

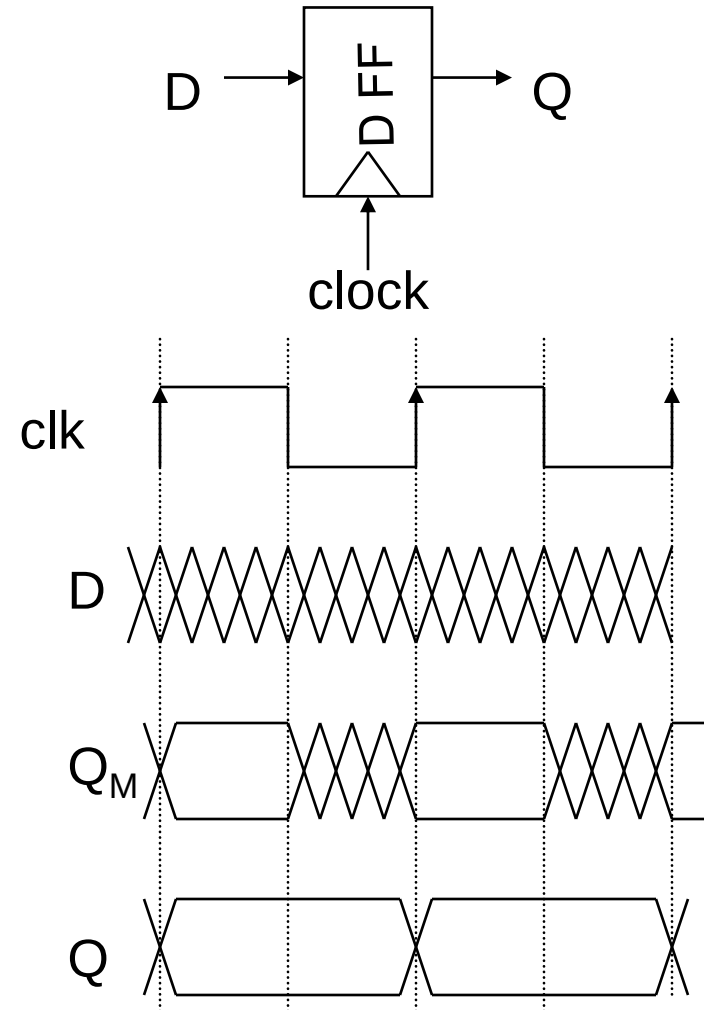


clk = 0 transparent

hold

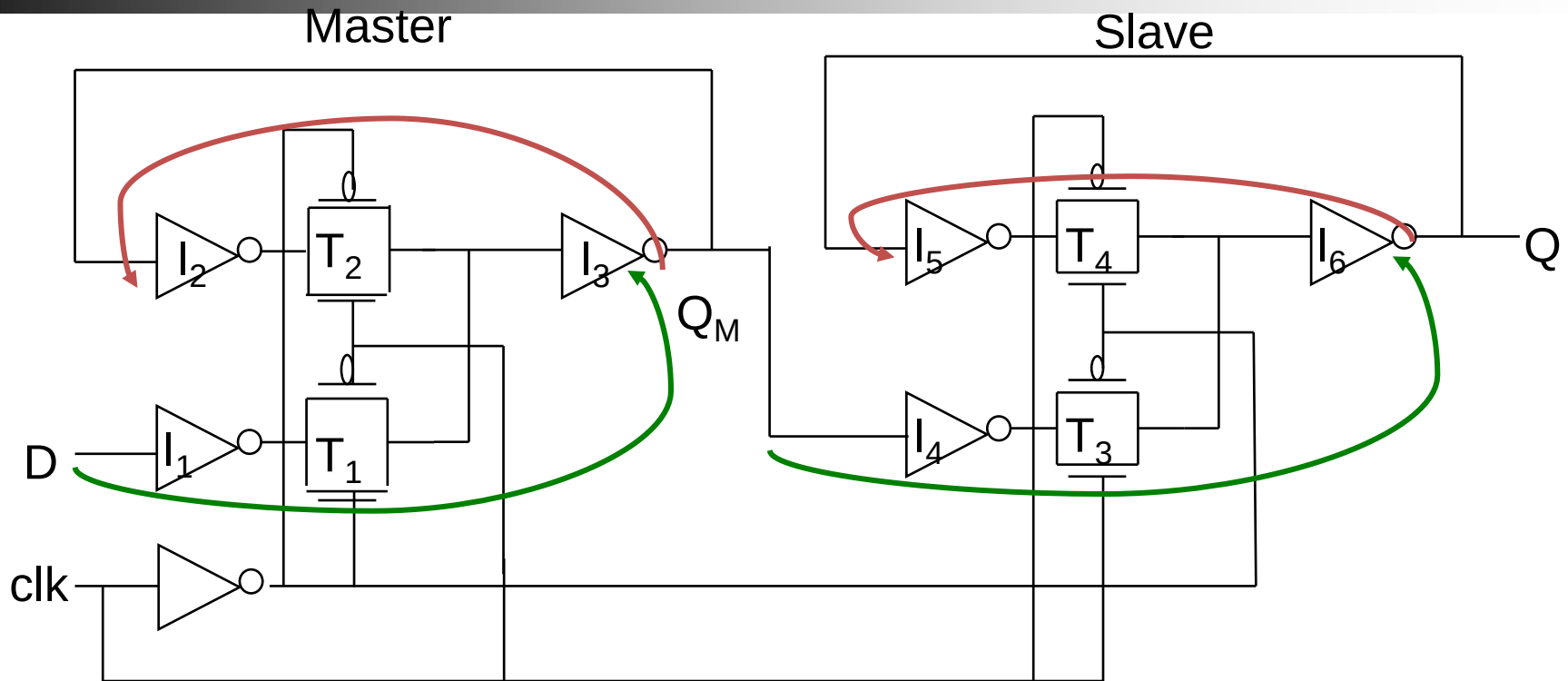
clk = 0 → 1 hold

transparent



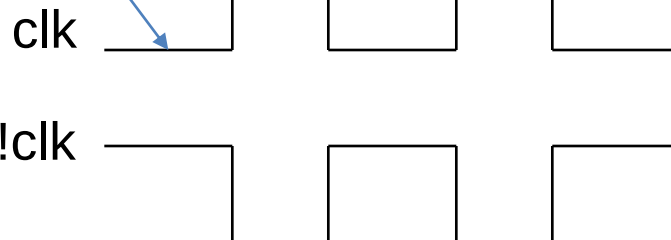


MS ET(edge triggered) Implementation



master transparent
slave hold

master hold
slave transparent



- t_{pd_inv} propagation delay of each inverter
- t_{pd_tx} propagation delay of transmis-gate



MS ET Timing Properties

- Assume propagation delays are t_{pd_inv} and t_{pd_tx} , that the contamination delay is 0, and that the inverter delay to derive !clk is 0
- **Set-up time** - time before rising edge of clk that D must be valid

$$3 * t_{pd_inv} + t_{pd_tx}$$

- **Propagation delay** - time for Q_M to reach Q

$$t_{pd_inv} + t_{pd_tx}$$

- **Hold time** - time D must be stable after rising edge of clk

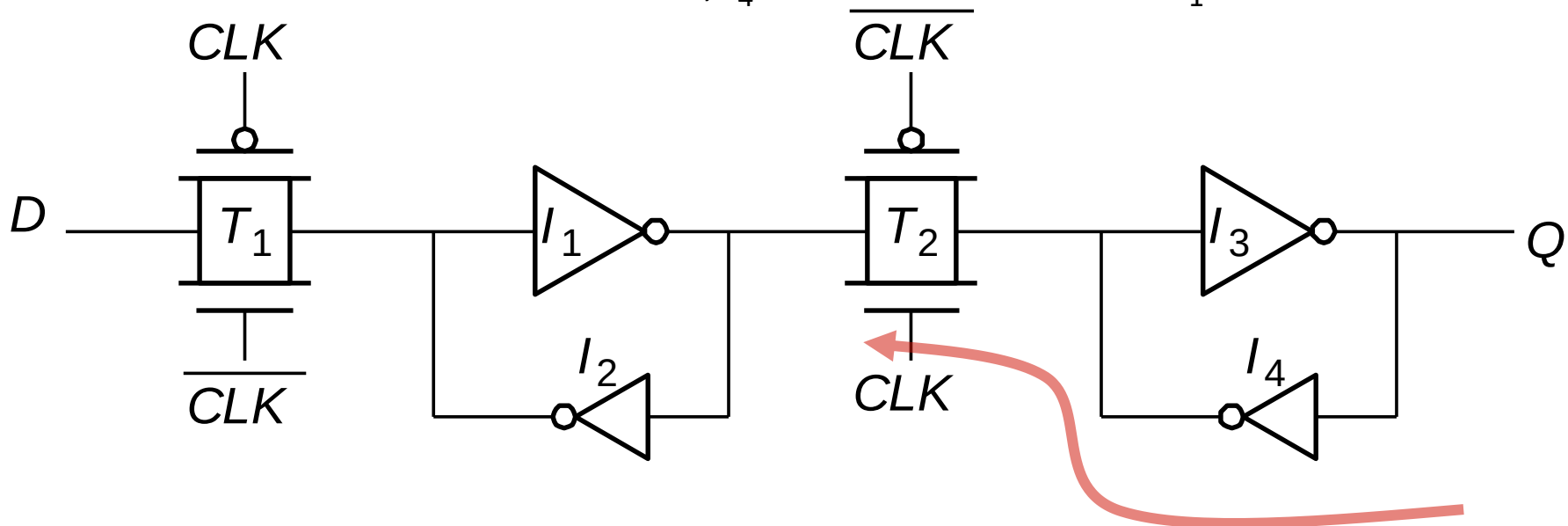
zero
Digital IC Design



Reduced Clock Load Master-Slave Register

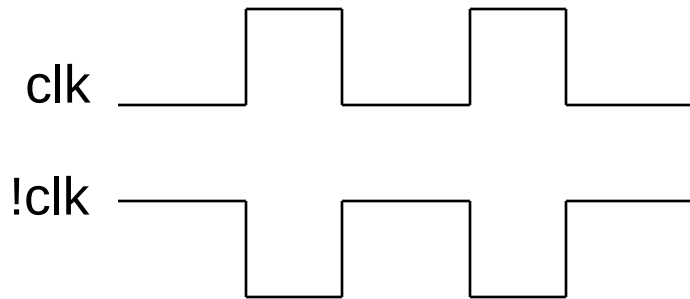
- Clock load per register is important since it directly impacts the power dissipation of the clock network.
- Can reduce the clock load (at the cost of robustness) by making the circuit ratioed

to switch the state of the master, T_1 must be sized to overpower I_2
to avoid reverse conduction, I_4 must be weaker than I_1

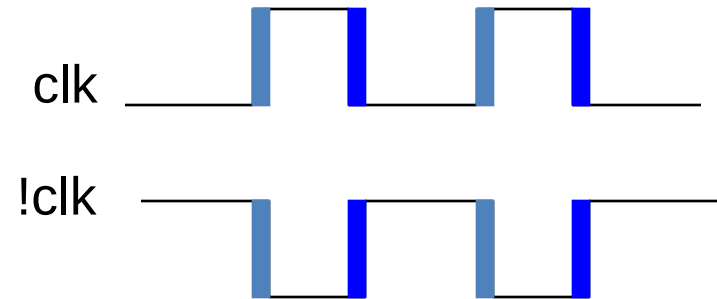




Non-Ideal Clocks



Ideal clocks



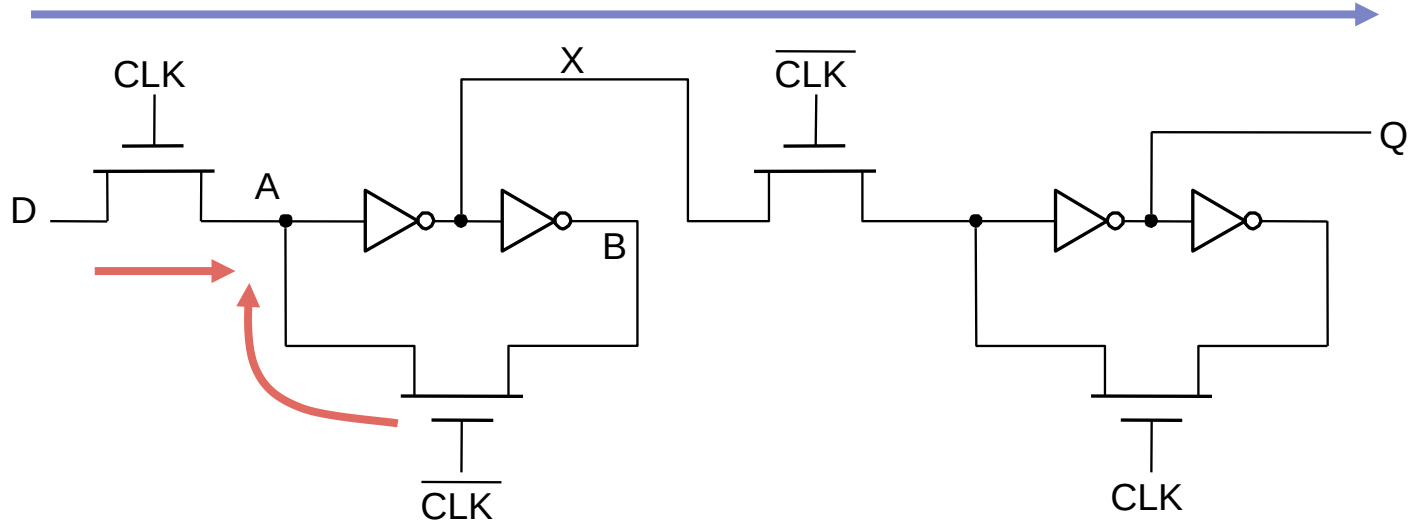
Non-ideal clocks
clock skew

1-1 overlap

0-0 overlap



Avoiding Clock Overlap (Clock Skew)

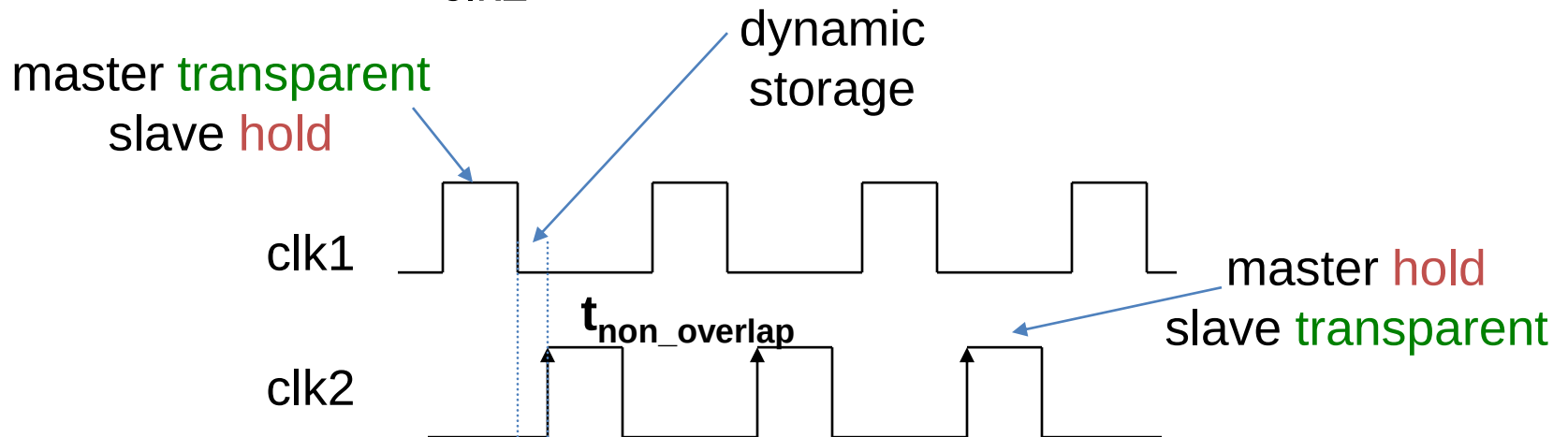
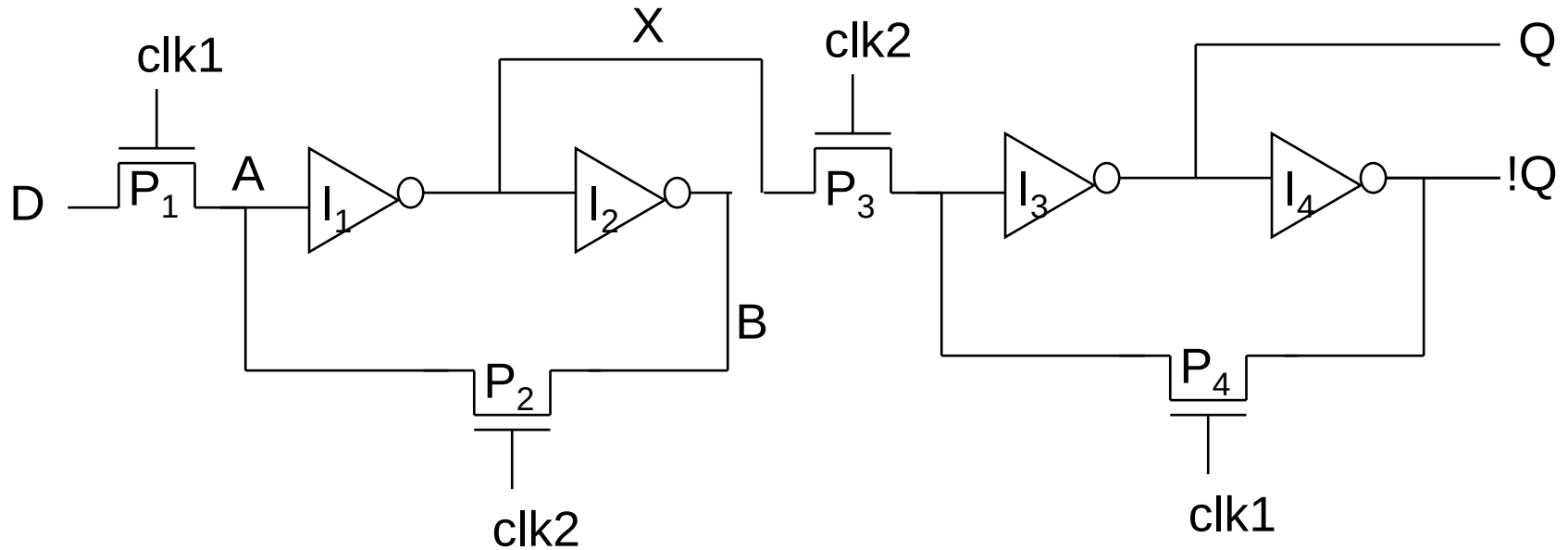


Race condition – direct path from D to Q during the short time when both clk and !clk are high (1-1 overlap)

Undefined state – both B and D are driving A when clk and !clk are both high

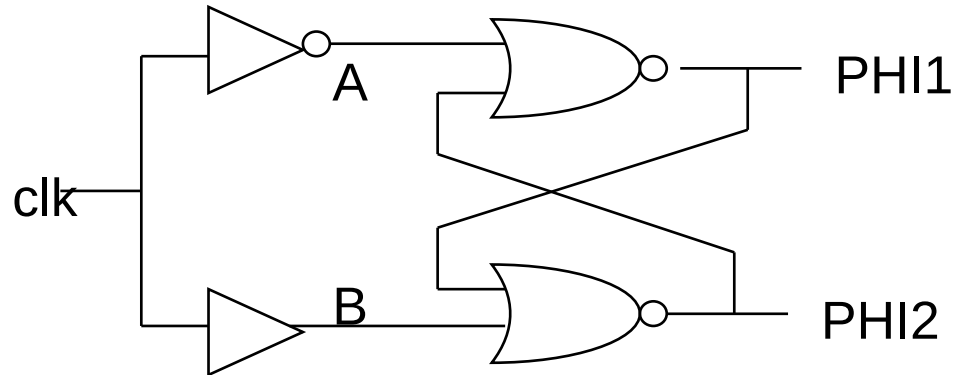


Pseudostatic Two-Phase ET FF



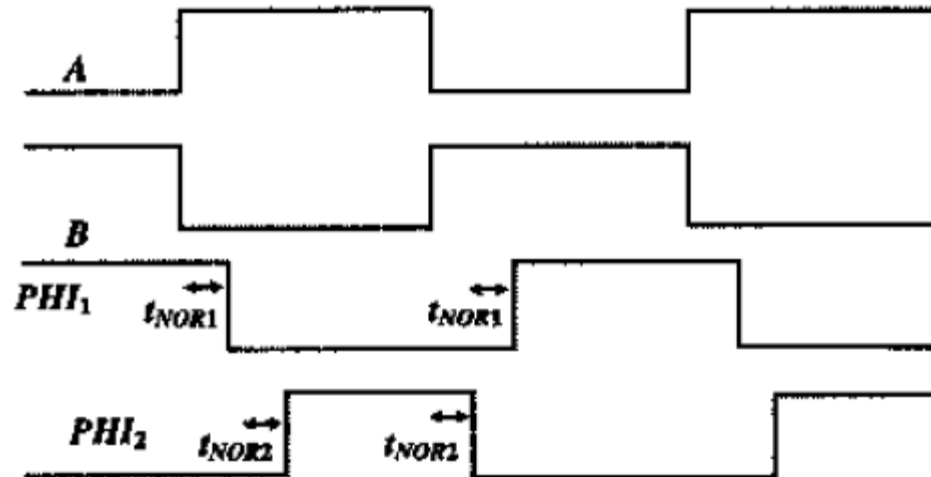


Two Phase Clock Generator



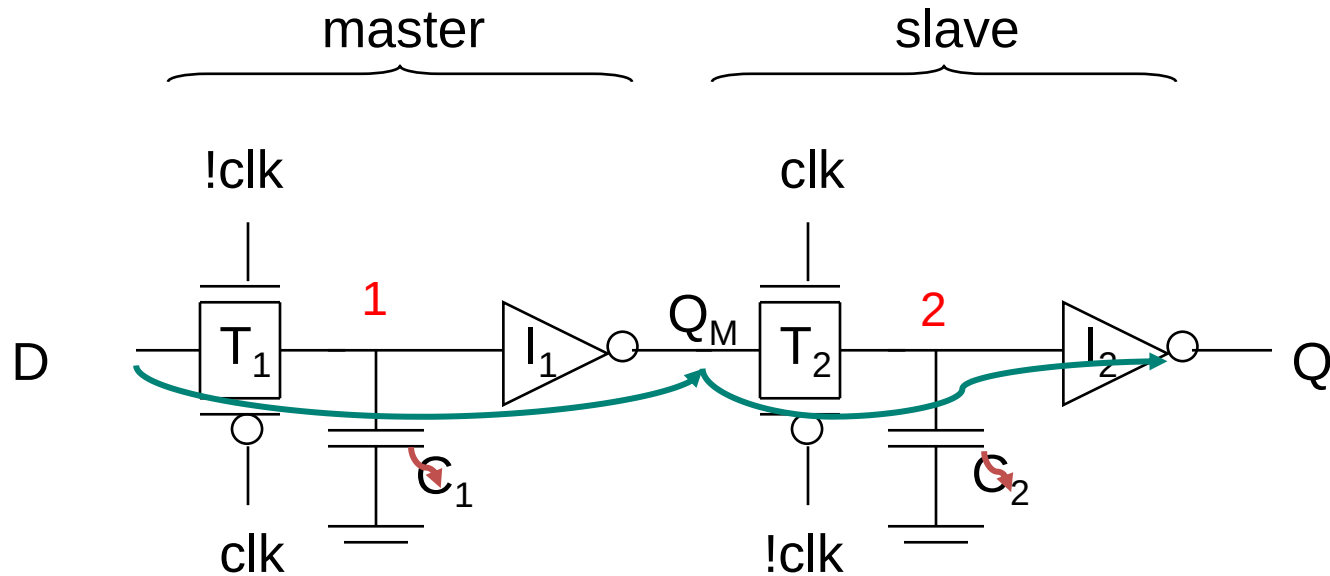
NOR Gate

A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

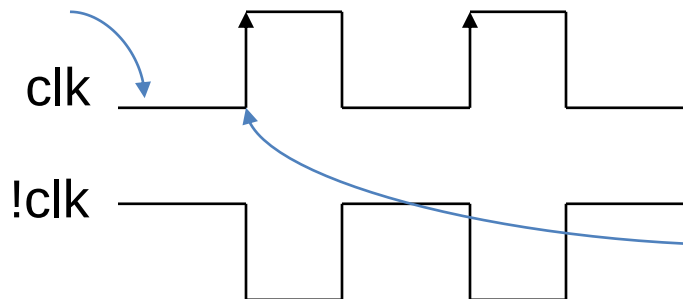




Dynamic ET Flipflop



master transparent
slave hold



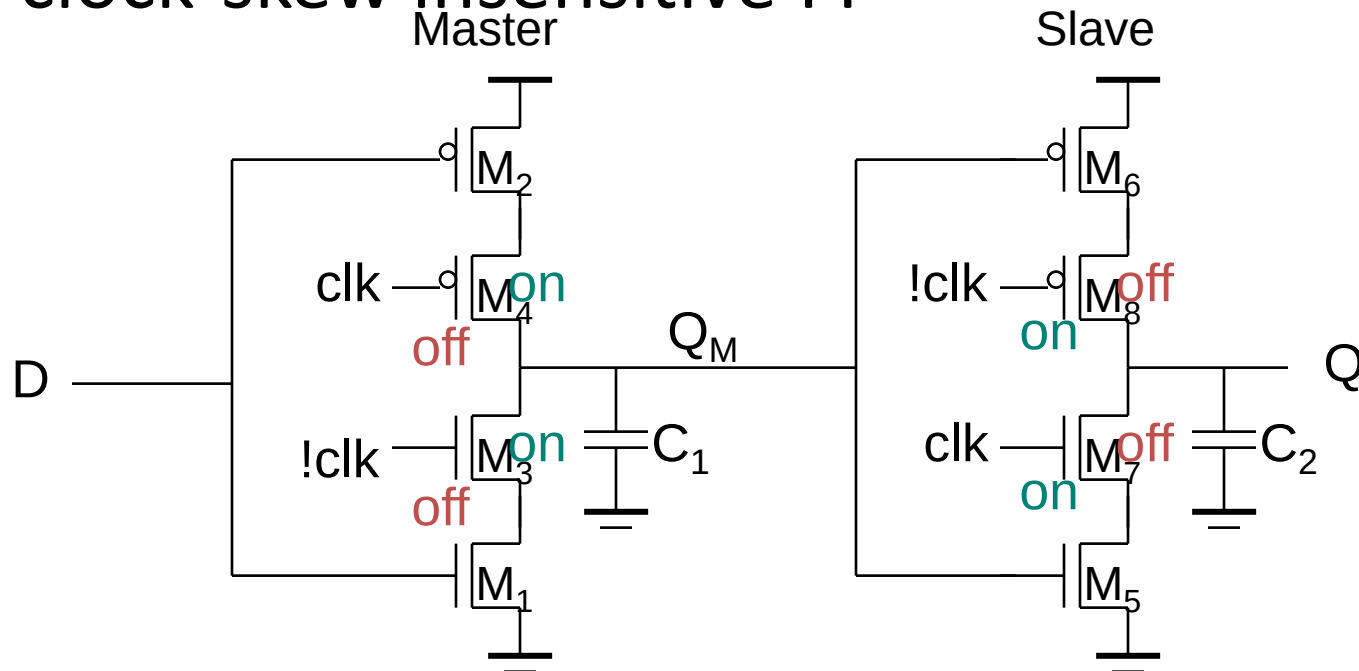
master hold
slave transparent

$$\begin{aligned} t_{su} &= t_{pd_tx} \\ t_{hold} &= \text{zero} \\ t_{c-q} &= 2 t_{pd_inv} + t_{pd_tx} \end{aligned}$$

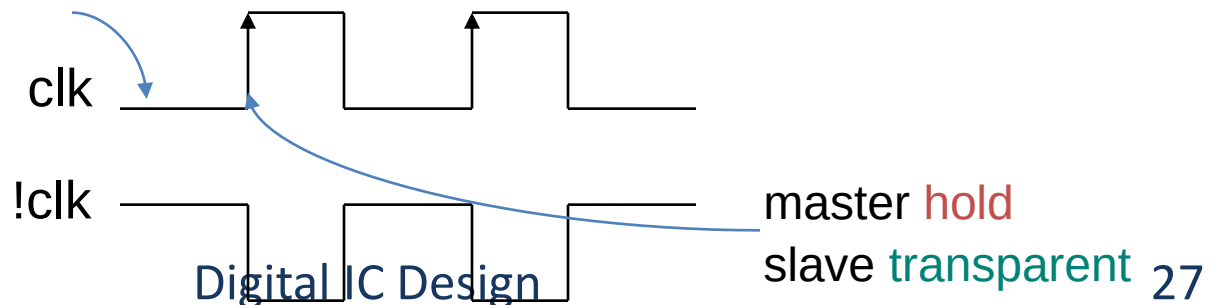


C²MOS (Clocked CMOS) ET Flipflop

- A clock-skew insensitive FF



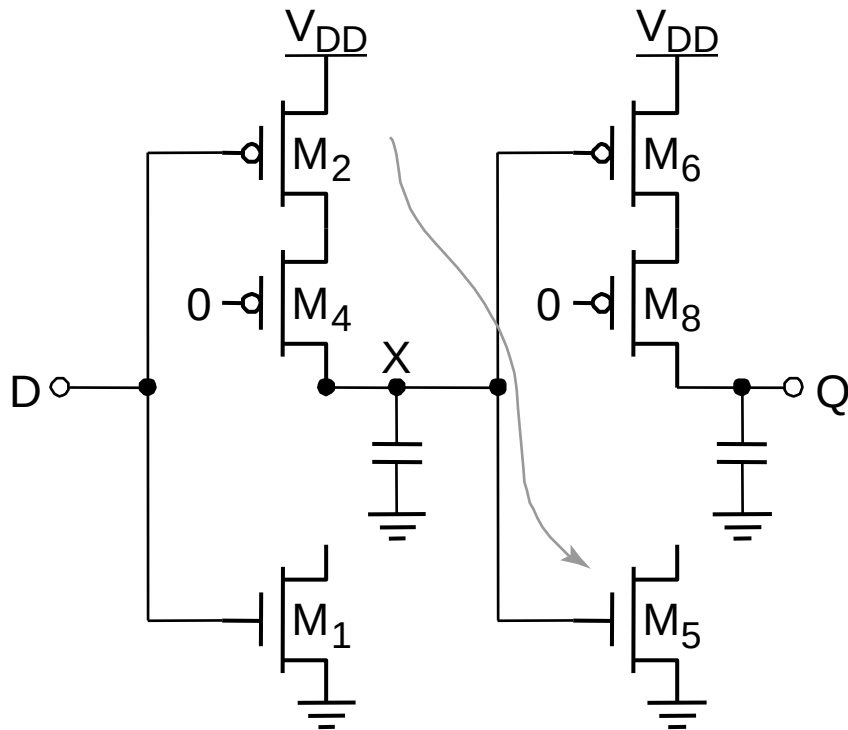
master transparent
slave hold



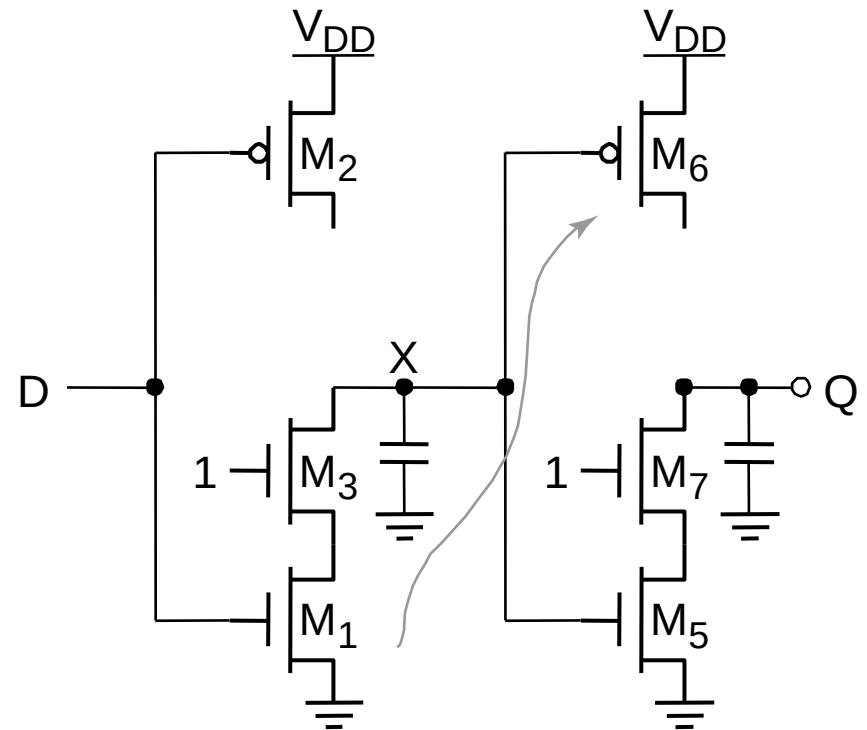
master hold
slave transparent



Insensitive to Clock-Overlap



(a) (0-0) overlap

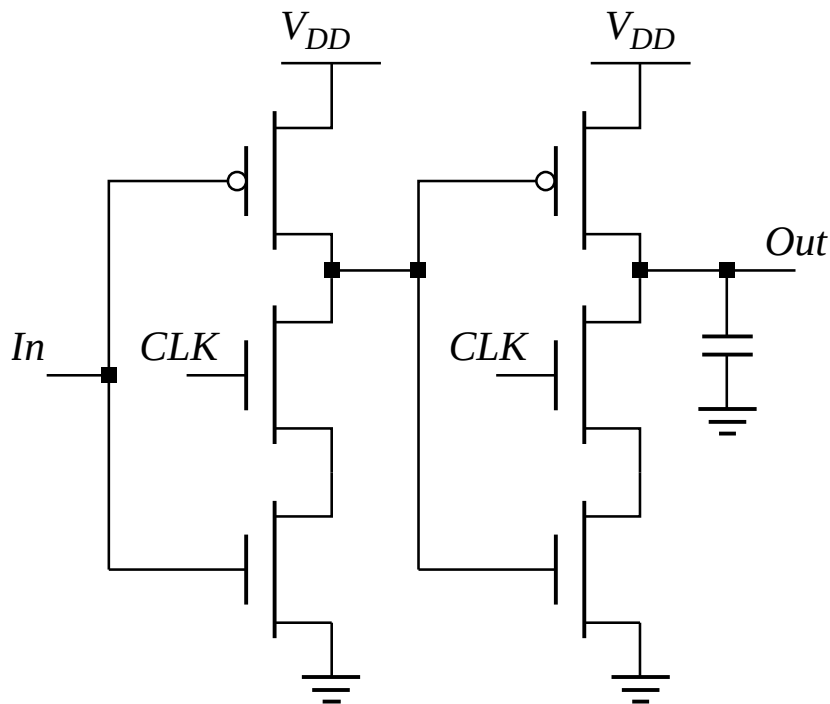


(b) (1-1) overlap

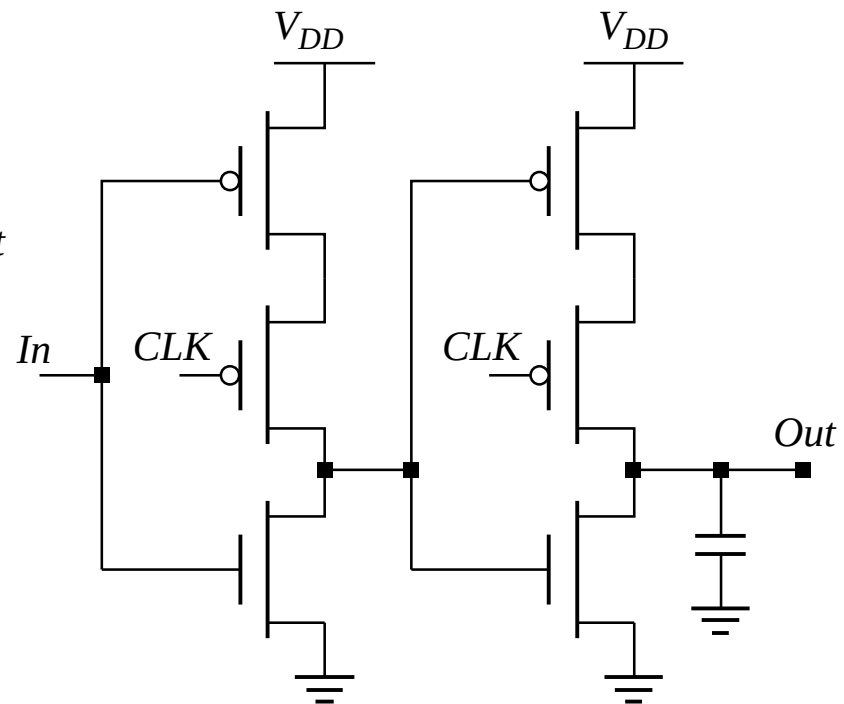


Other Latches/Registers:

True Single-Phase Clocked Register (TSPCR)



Positive latch
(transparent when CLK=1)

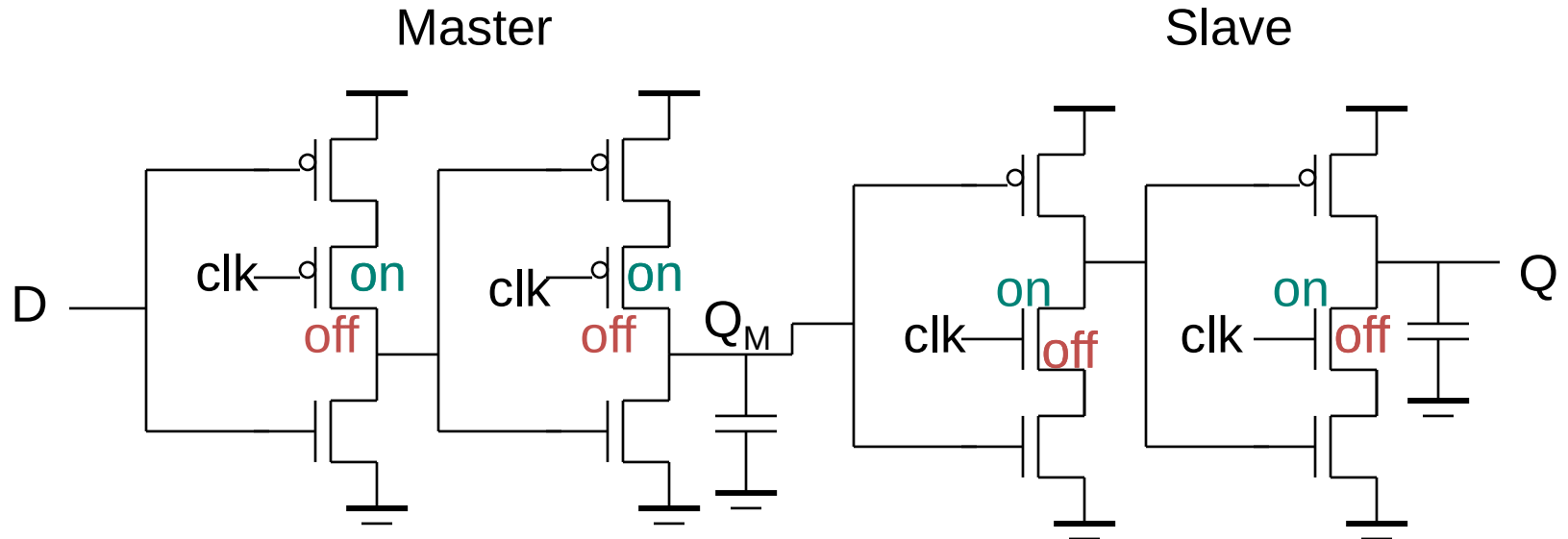


Negative latch
(transparent when CLK=0)

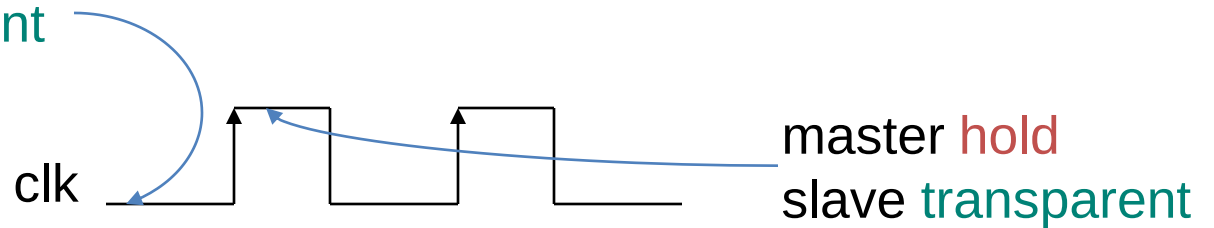
TSPC: True Single-Phase Clocked Register [Yuan89]
Digital IC Design



TSPC ET FF

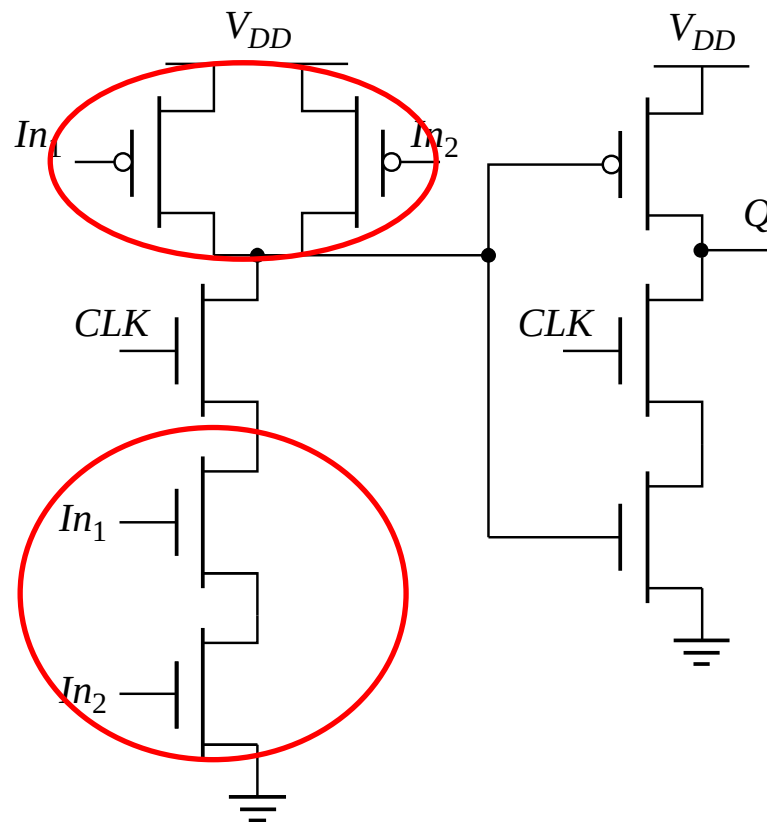
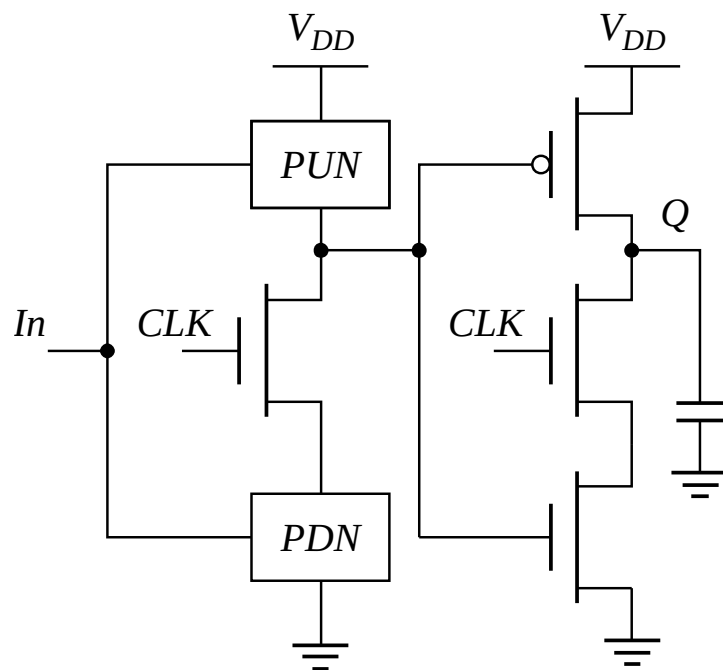


master transparent
slave hold





Including Logic in TSPCR



Example: logic inside the latch
For Specific high performance CPU DP

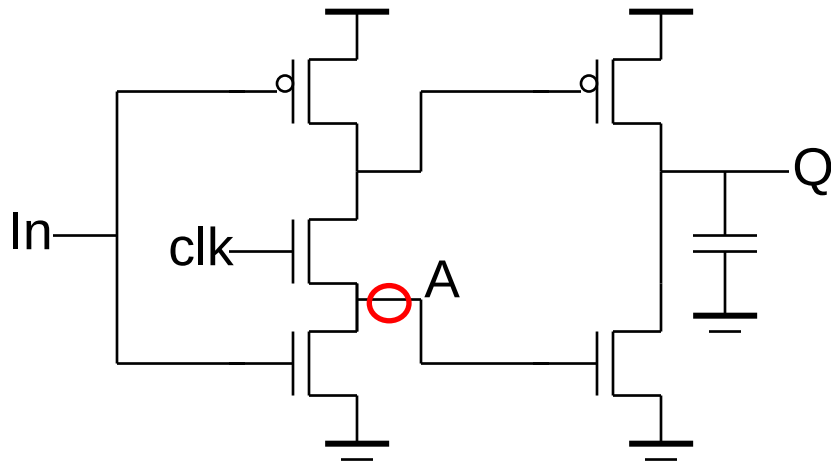
AND latch
AND+latch

Setup T=140ps
Setup T=600ps



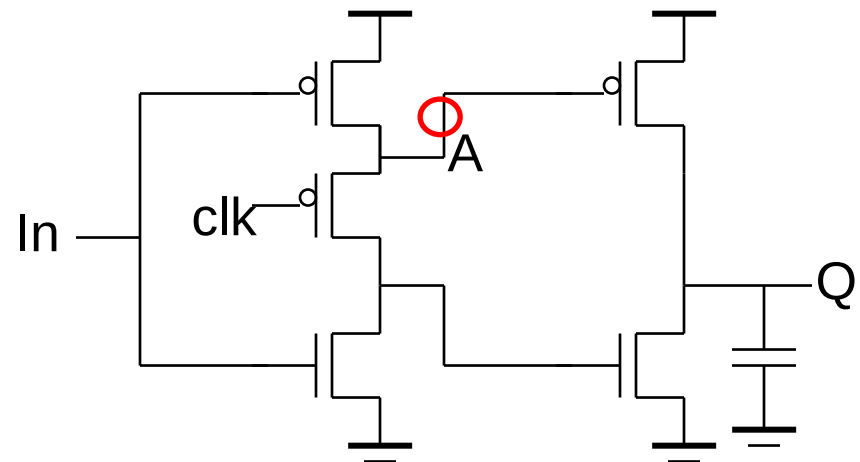
Split-Output TSPC Latches

Positive Latch



transparent when $\text{clk} = 1$
hold when $\text{clk} = 0$

Negative Latch



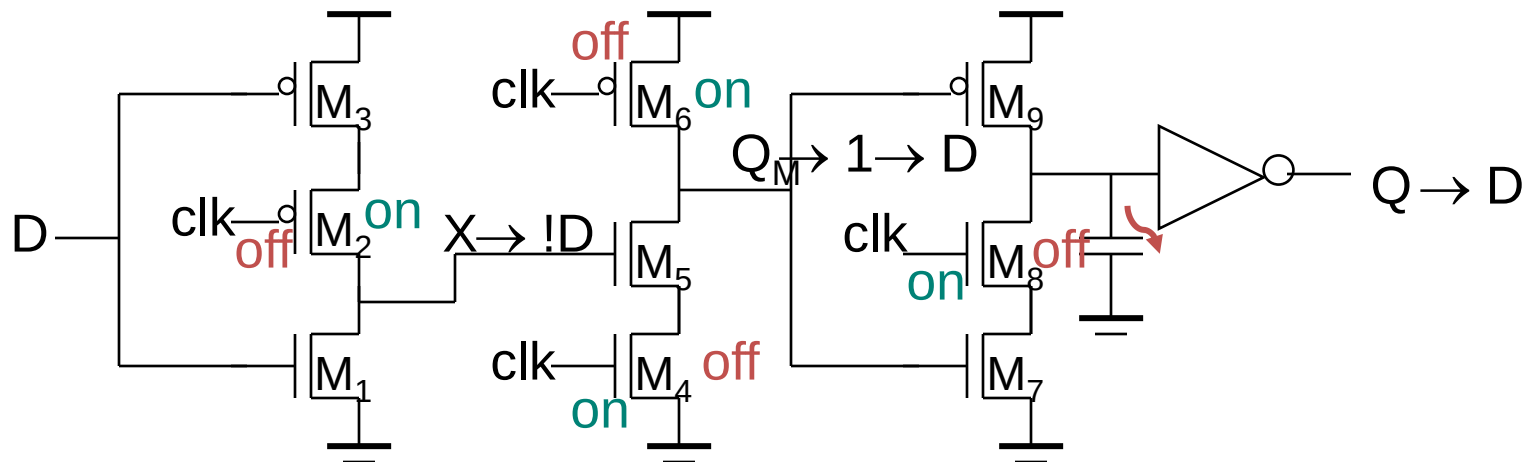
hold when $\text{clk} = 1$
transparent when $\text{clk} = 0$

When $\text{In} = 0$, $A_{\text{max}} = V_{\text{DD}} - V_{\text{Tn}}$

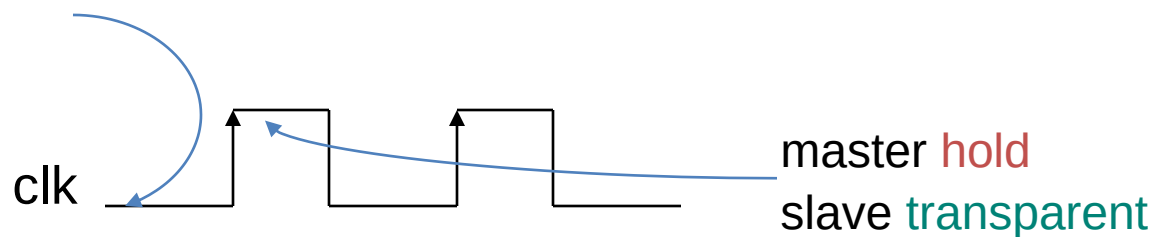
When $\text{In} = 1$, $A_{\text{min}} = |V_{\text{Tp}}|$



Simplified TSPCR ET FF

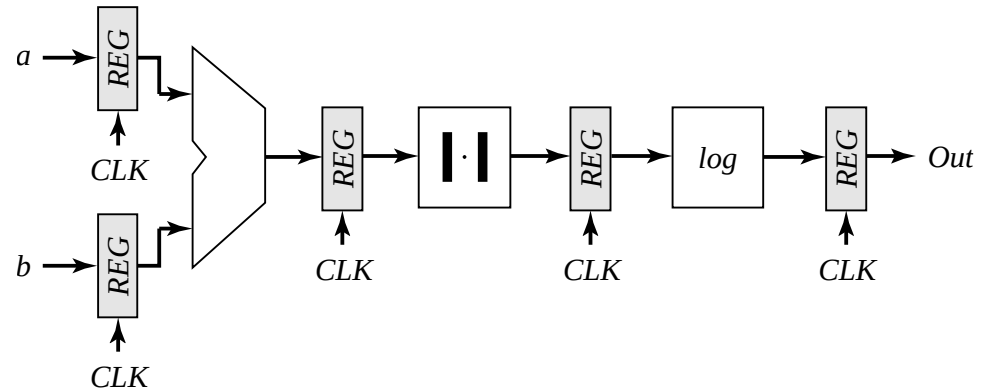
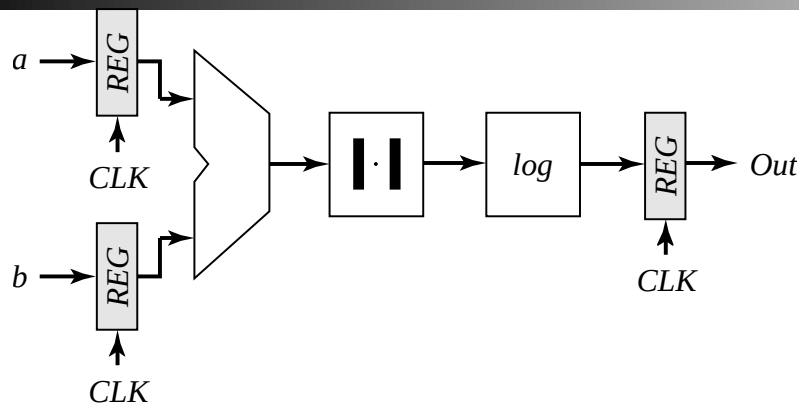


master transparent
slave hold





Pipelining



Conventional: The minimal clock period $T_{min} = t_{c-q} + t_{pd,logic} + t_{su}$

Pipeline: $T_{min,pipe} = t_{c-q} + \max(t_{pd,add}, t_{pd,abs}, t_{pd,log})$

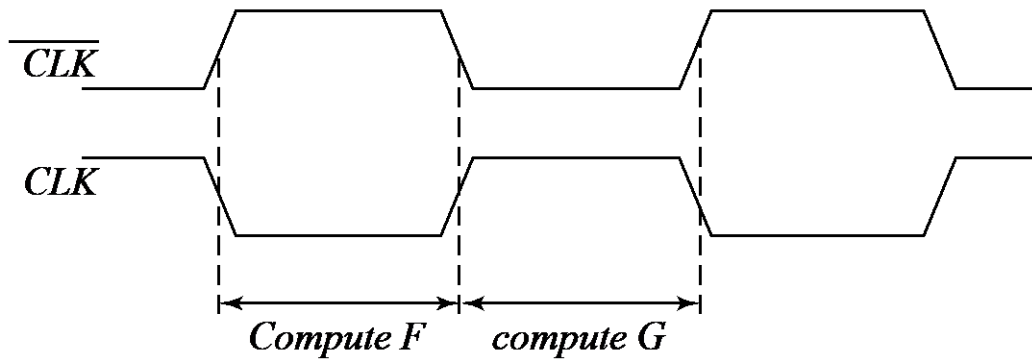
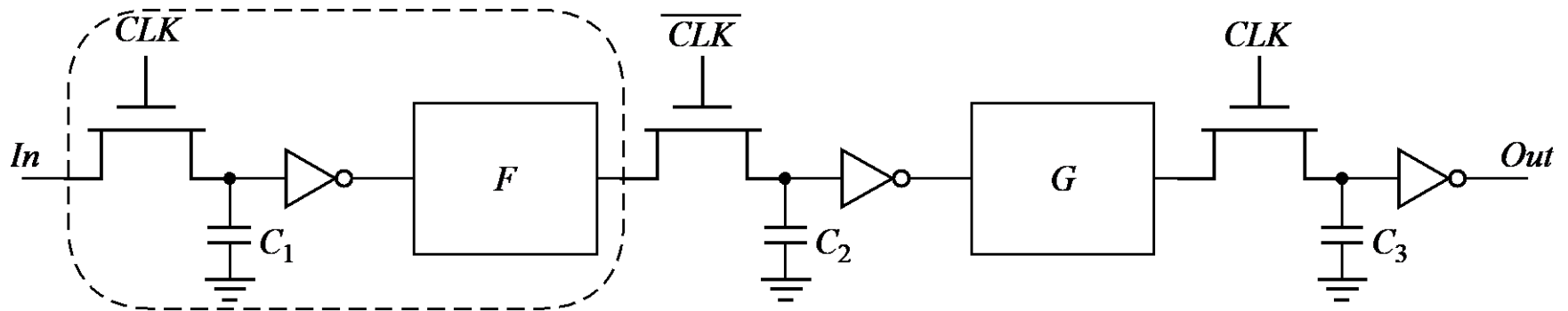
Reference

Clock Period	Adder	Absolute Value	Logarithm
1	$a_1 + b_1$		
2	$a_2 + b_2$	$ a_1 + b_1 $	
3	$a_3 + b_3$	$ a_2 + b_2 $	$\log(a_1 + b_1)$
4	$a_4 + b_4$	$ a_3 + b_3 $	$\log(a_2 + b_2)$
5	$a_5 + b_5$	$ a_4 + b_4 $	$\log(a_3 + b_3)$

Pipelined

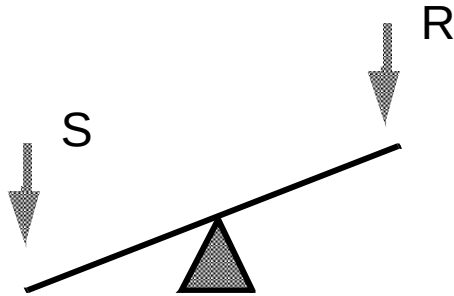


Latch-Based Pipeline

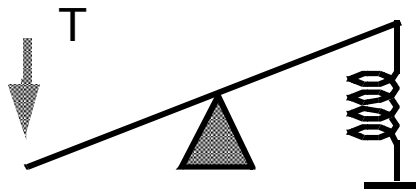




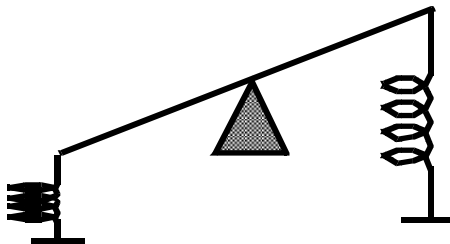
Multivibrator Circuits



Bistable Multivibrator
flip-flop, Schmitt Trigger



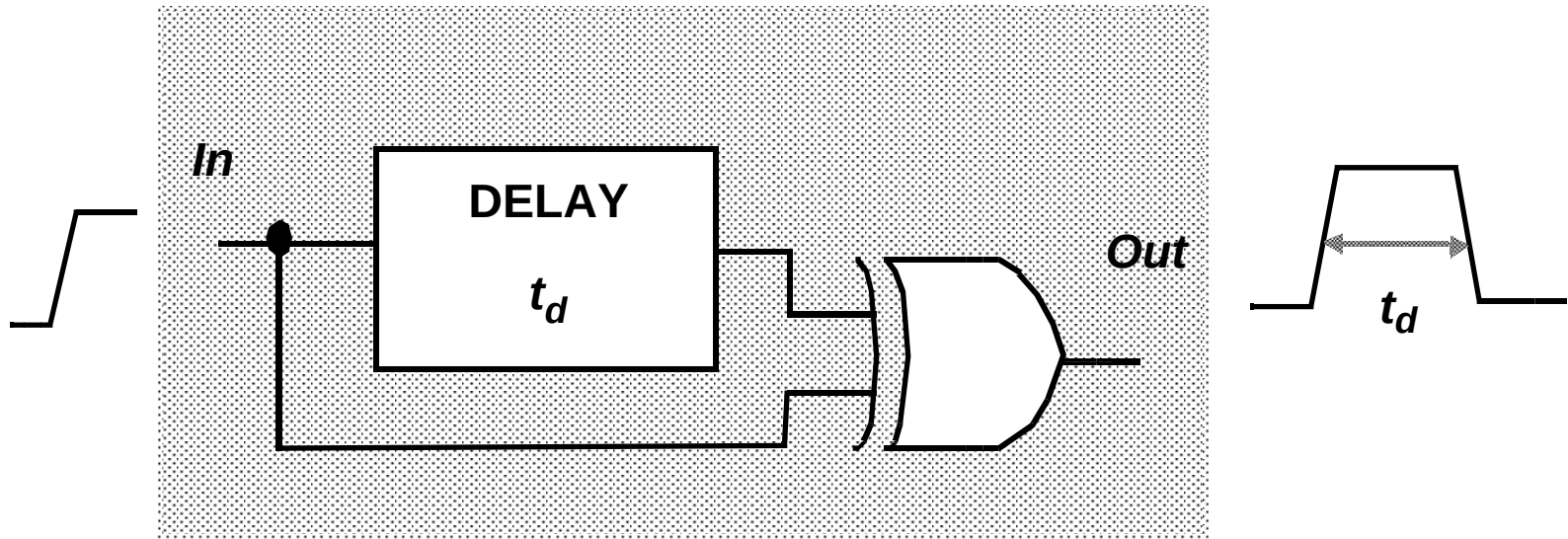
Monostable Multivibrator
one-shot



Astable Multivibrator
oscillator

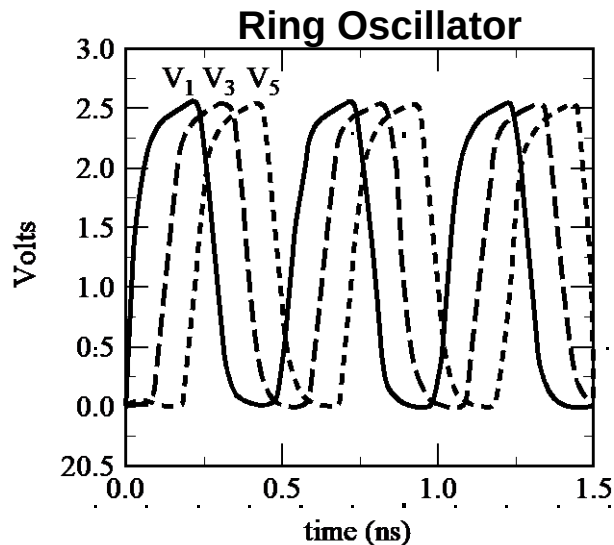
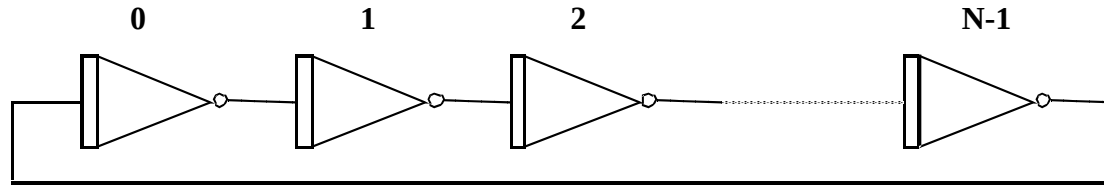


Transition-Triggered Monostable





Astable Multivibrators (Oscillators)



simulated response of 5-stage oscillator

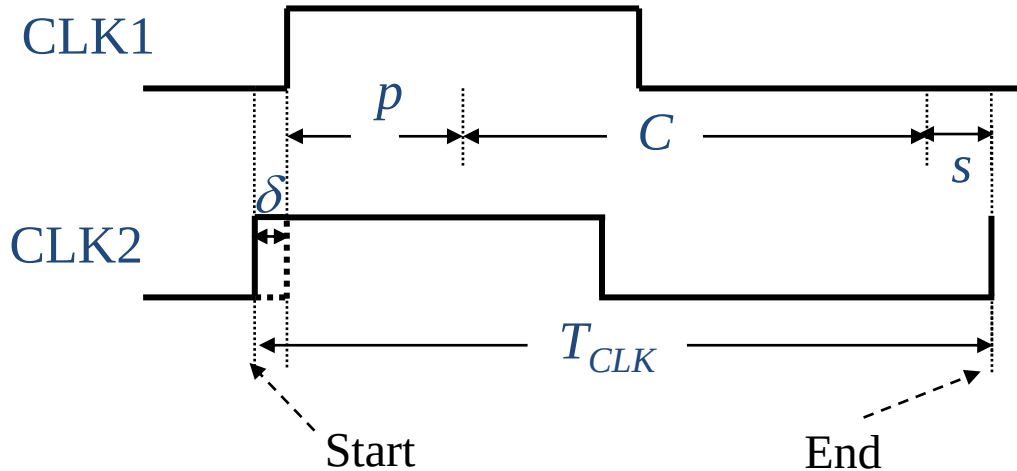
$$T_{osc} = 2 * t_p * N,$$

Digital IC Design



Setup Time Constraint

Clock Period Constraint

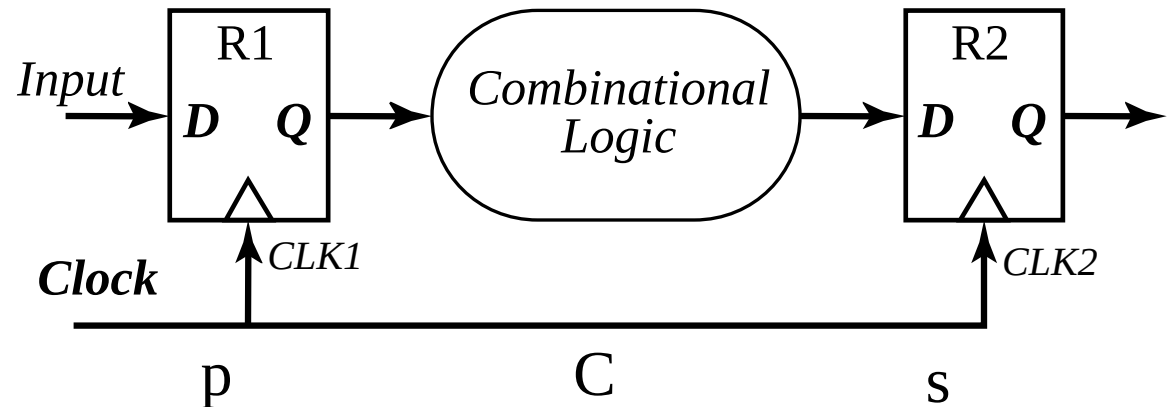


Minimum Cycle Time:

$$T_{CLK} - \delta \geq p + s + C_{max}$$

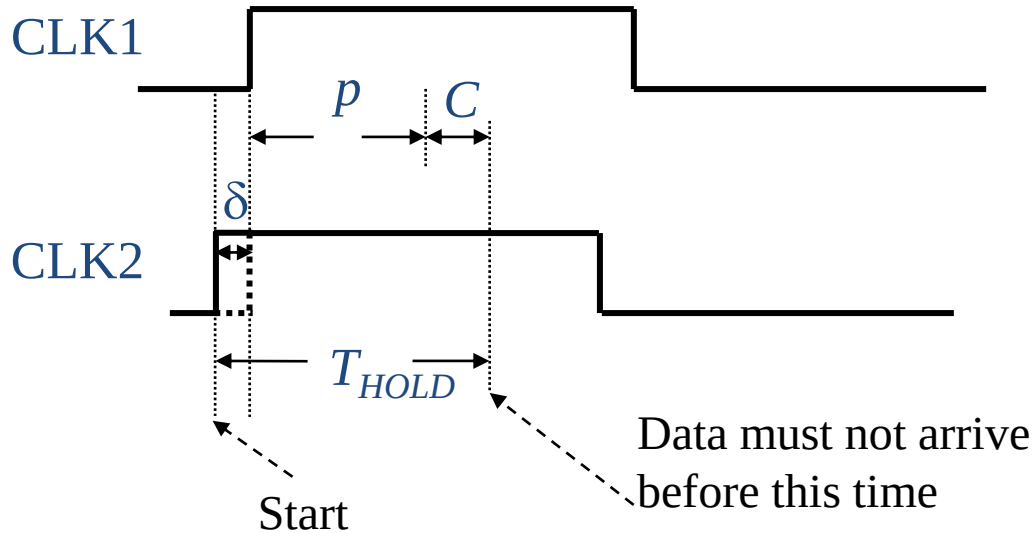
Worst case: rising edge of CLK2 arrives early (positive δ)

The Longest Path determines the setup time constraint!





Hold Time Constraint

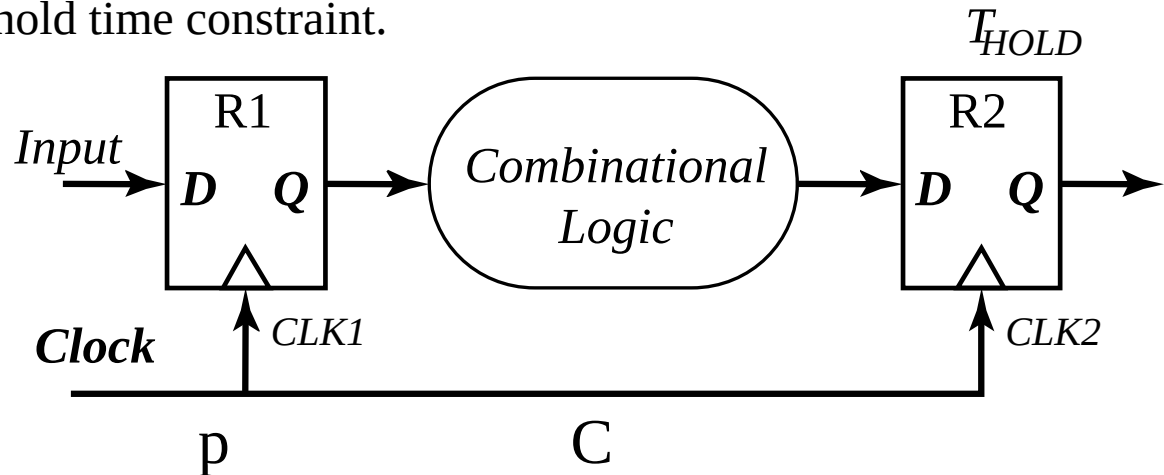


Hold Time Constraint:

$$T_{HOLD} - \delta < p + C_{min}$$

Worst case is when rising edge of CLK2 arrives late (negative δ)

The shortest Path determines the hold time constraint.





Choosing a Clocking Strategy

- Choosing the right clocking scheme affects the functionality, speed, and power of a circuit
- Two-phase designs
 - + robust and conceptually simple
 - - need to generate and route two clock signals
 - - have to design to accommodate possible skew between the two clock signals
- Single phase designs
 - + only need to generate and route one clock signal
 - + supported by most automated design methodologies
 - + don't have to worry about skew between the two clocks
 - - have to have guaranteed slopes on the clock edges



Summary

- The cross-coupling of two inverters creates a bistable circuit
- A latch is a level-sensitive memory element that samples data on one phase and holds data on the other phase
- A register on the other hand samples the data on the rising or falling edge, master-slave configuration
- Registers can be static or dynamic
- Monostable, Astable